

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

*Design and Implementation of a Mobile App
for Collection of User Experience Data from
Mobile Network Subscribers*

GROUP 19

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the requirements for the development of a mobile application that collects Quality of Experience (QoE) data from mobile network subscribers in real time. This document provides a detailed description of the system functionalities, features, constraints, and expected behavior.

The SRS document serves as a communication tool between:

- Developers
- Stakeholders
- Testers
- Project managers
- Mobile network operators

It ensures that all parties clearly understand the objectives and requirements of the proposed system.

1.2 Scope

The proposed mobile application is designed to collect and analyze QoE data from mobile subscribers in Cameroon. The system combines:

- Subjective user feedback
- Objective network performance metrics

The application will:

- Run continuously in the background
- Monitor network performance automatically
- Prompt users for satisfaction ratings
- Store and synchronize collected data
- Generate reports and analytics
- Passively collect objective network performance data (signal strength, latency, speed, GPS location) in the background
- Actively collect subjective QoE feedback from users through periodic prompts
- Store data locally when offline and sync to the backend server when connected
- Display real-time network statistics and weekly experience summaries
- Support both English and French languages
- Help mobile network operators in Cameroon understand and improve network quality

The collected information will help:

- Improve mobile network performance

- Identify network issues
- Support decision-making by telecom operators
- Enhance customer satisfaction

1.3 Objectives

The objectives of the system are:

- To collect real-time QoE data from subscribers
- To monitor mobile network performance continuously
- To combine subjective feedback with objective measurements
- To support QoE analytics and reporting
- To improve understanding of user experience in Cameroon
- To provide reliable data for network optimization

1.4 Definitions and Acronyms

Acronyms	Meaning
QoE	Quality of Experience : the user's perception of service quality
QoS	Quality of Service : technical network performance metrics
GPS	Global Positioning System
API	Application Programming Interface
UI	User Interface
DBMS	Database Management System
KPI	Key Performance Indicator
FR	Functional Requirement: what the system does
NFR	Non-Functional Requirement: how the system performs
RSRP	Reference Signal Received Power: a signal strength metric
RSSI	Received Signal Strength Indicator
LTE	Long Term Evolution: 4G mobile network standard
AES-256	Advanced Encryption Standard (256-bit)
HTTPS/TLS	Secure data transmission protocols
SRS	Software Requirements Specification: this document

2. Overall Description

2.1 Product Perspective

The system is a mobile application that works alongside existing mobile network services. The application continuously monitors network and device metrics while collecting user feedback regarding network quality and satisfaction. The system is designed to complement existing network monitoring tools by incorporating real user experiences into network analysis.

Existing platforms analyzed during requirement gathering include:

- Speedtest by Ookla
- OpenSignal
- NetMonster

These systems inspired certain functionalities but do not fully combine automated QoE monitoring with user satisfaction collection in the proposed way.

2.2 Product Functions

The main functions of the application include:

- User registration and authentication
- Continuous background monitoring
- Collection of network performance metrics
- User feedback collection
- GPS location tracking
- Timestamp logging
- Data synchronization with backend server
- QoE report generation
- Network analytics visualization

The system should also:

- Runs in the background collecting network metrics automatically
- Prompts users once per day for satisfaction ratings
- Stores all data locally when the device is offline
- Syncs data to the backend server when internet is restored
- Displays real-time network statistics on a dashboard
- Allows users to manually report network problems
- Generates weekly in-app network experience summaries

2.3 User Classes and Characteristics

Primary Users

- Mobile network subscribers
- General smartphone users

Secondary Users

- Network analysts
- Researchers
- Telecom operators

Administrators

- Manage reports
- Manage system configurations
- Analyze collected data

User Class	Role
Mobile Subscriber	Primary user: installs app, provides feedback, views personal network status
Network Operator	Secondary: accesses backend dashboard to analyze aggregated data
System Administrator	Manages backend server and database

2.4 Operating Environment

The system will operate in the following environment:

Hardware Requirements

- Android smartphone
- GPS-enabled device
- Internet connectivity

Software Requirements

- Android operating system
- Android Studio
- Java/Kotlin
- Firebase or MySQL database

- Backend server (Node.js)

Parameter	Specification
Platform	Android 8.0 (API Level 26) and above
Minimum RAM	2 GB
Minimum Storage	50 MB available
Network	Works on 2G, 3G, 4G/LTE and Wi-Fi
Backend	RESTful API server with cloud database
Languages	English and French

2.5 Design and Implementation Constraints

The system has the following constraints:

- Internet access is required for data synchronization
- Android permissions are necessary for location and network monitoring
- Battery optimization restrictions may affect background services
- Some network metrics depend on device compatibility
- Comply with data protection regulations in Cameroon
- Must not require constant internet connectivity to operate
- Must function in low-connectivity environments common in Cameroon

2.6 Assumptions and Dependencies

The following assumptions were made:

- Users will allow required permissions
- Users will provide honest feedback
- Telecom operators may collaborate in the future
- Users will keep internet connectivity enabled

3. Functional Requirements

3.1 User Authentication

The system must:

- Allow user registration
- Allow secure login and logout
- Store user credentials securely

3.2 Background Monitoring

The application must:

- Run continuously in the background
- Collect data periodically
- Minimize battery consumption

3.3 Network Metrics Collection

The application must collect:

- Signal strength
- Network type (2G, 3G, 4G, 5G)
- Download speed
- Upload speed
- Latency (Ping)
- Jitter
- Packet loss
- Bandwidth

3.4 User Feedback Collection

The system must:

- Prompt users for satisfaction ratings
- Allow users to submit comments
- Record usability feedback

3.5 GPS and Timestamp Logging

The application must:

- Record GPS location
- Record timestamps for all collected data

3.6 Data Synchronization

The system must:

- Send collected data to the backend server
- Synchronize automatically when internet becomes available

3.7 Report Generation

The system must:

- Generate QoE analytics reports
- Display network performance trends
- Visualize collected data using charts and graphs

Functional Requirements Classification.

FR01 — Background Network Monitoring | Priority: Must Have

Description	The app shall continuously monitor network quality in the background without requiring the user to keep the app open.	
I/O	Input: Device network APIs	Output: Periodic network metric logs stored in local database

FR02 Real-Time Statistics Display | Priority: Should Have

Description	The app shall display a dashboard showing current network statistics including signal strength, latency, speed, and network type.	
I/O	Input: Live collected metrics	Output: Dashboard screen with updated statistics

FR03 — Network Drop Notifications | Priority: Should Have

Description	The app shall notify the user when download speed drops below 1 Mbps or signal strength drops below -100 dBm.
I/O	Input: Real-time metrics Output: Push notification on the device

FR04 — Signal Strength Collection | Priority: Must Have

Description	The app shall collect and record RSRP and RSSI signal strength metrics every 60 seconds.
I/O	Input: Device Telephony Manager API Output: Signal values stored with timestamp

FR05 — Latency and Ping Collection | Priority: Must Have

Description	The app shall measure and record network latency (ping) values periodically.
I/O	Input: Network ping requests Output: Latency values in milliseconds stored in database

FR06 — Download and Upload Speed Collection | Priority: Must Have

Description	The app shall measure and record download and upload speeds at periodic intervals using lightweight requests.
I/O	Input: Network speed test calls Output: Speed values in Mbps stored in database

FR07 — GPS Location Tagging | Priority: Must Have

Description	The app shall tag every measurement with GPS coordinates, timestamp, and location name.
I/O	Input: Device GPS sensor Output: Location-tagged records in database

FR08 — Offline Data Storage | Priority: Must Have

Description	The app shall store all collected data locally using SQLite for up to 7 days when no internet is available.	
I/O	Input: Collected metrics and feedback	Output: Local SQLite database records

FR09 — Backend Data Synchronization | Priority: Must Have

Description	The app shall automatically sync all pending local data to the backend server immediately when internet connectivity is restored.	
I/O	Input: Local SQLite records	Output: Data uploaded to backend REST API

FR10 — Network Quality Map | Priority: Could Have

Description	The app shall display an interactive map showing crowdsourced network quality levels across different locations.	
I/O	Input: Aggregated backend data	Output: Color-coded map visualization

FR11 — User Feedback Prompts | Priority: Must Have

Description	The app shall prompt the user once per day to rate their network experience on a scale of 1 to 5. The prompt shall be dismissible.	
I/O	Input: User rating (1–5)	Output: Rating stored with timestamp and GPS location

FR12 — Manual Problem Reporting | Priority: Should Have

Description	The app shall allow users to manually submit a network problem report including description, location, and severity.	
I/O	Input: User text and severity selection	Output: Report stored and synced to backend

FR13 — Weekly Experience Reports | Priority: Could Have

Description	The app shall generate a weekly in-app summary of the user's average network quality and most common problems.	
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I/O	Input: Last 7 days of stored data Output: Weekly summary screen
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FR14 — Subjective QoE Feedback Collection | Priority: Must Have

Description	The app shall link user satisfaction ratings directly to the objective network metrics collected at the same time and location.
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I/O	Input: User ratings from FR11 Output: Linked QoE and objective records in database
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FR15 — Bilingual Support | Priority: Must Have

Description	The app shall support English and French throughout the entire interface, with a language toggle in settings.
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I/O	Input: Language selection Output: Full UI rendered in selected language
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FR16 — Data Sharing Control | Priority: Must Have

Description	The app shall allow users to view what data is collected, control sharing preferences, and opt out at any time.
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I/O	Input: User privacy settings Output: Updated data collection behaviour
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4. Non-Functional Requirements

Performance

- The application must respond quickly
- Data collection must not slow down the device

Reliability

- The system must run continuously without crashing
- Data loss should be minimized

Security

- User data must be encrypted
- Authentication mechanisms must be secure

Usability

- The interface must be user-friendly
- Navigation must be simple

Scalability

- The system must support thousands of users simultaneously

Maintainability

- The application should be easy to update and maintain

Non Functional Requirements Specification.

NFR01 — Battery Optimization | Priority: Should Have

Description	The app shall consume no more than 3–5% of device battery per day during background operation.
I/O	Category: Performance

NFR02 — Data Consumption Limit | Priority: Should Have

Description	The app shall consume no more than 5 MB of mobile data per day during normal background operation.
I/O	Category: Performance

NFR03 — Low-End Device Compatibility | Priority: Should Have

Description	The app shall function correctly on Android 8.0+ devices with minimum 2 GB RAM and 50 MB available storage.
I/O	Category: Compatibility

NFR04 — Data Security and Privacy | Priority: Must Have

Description	All data stored locally shall be encrypted using AES-256. All data transmitted to the server shall use HTTPS with TLS 1.2 or higher.
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NFR08 — Backend Scalability | Priority: Could Have

Description	The backend shall handle data from at least 10,000 concurrent users without performance degradation.
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NFR05 — Application Response Time | Priority: Should Have

Description	The app shall load and display the main dashboard within 3 seconds of being opened.
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I/O	Category: Performance
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NFR06 — Offline Reliability | Priority: Must Have

Description	The app shall continue collecting and storing data locally for a minimum of 7 days without internet and without data loss.
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I/O	Category: Reliability
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NFR07 — Usability | Priority: Should Have

Description	Core functions shall be accessible within 2 taps from the home screen. The interface shall be usable by non-technical users without training.
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I/O	Category: Usability
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5. System Features

The application includes the following features:

- Real-time QoE monitoring
- Continuous background execution
- User satisfaction rating system
- Speedtest integration
- Analytics dashboard
- Interactive graphs

- Notification system
- Exportable reports
- GPS-based network analysis

6. External Interface Requirements

User Interface

The application interface must include:

- Login screen
- Main Dashboard: real-time network stats (signal, latency, speed, network type)
- Settings Screen: language selection and privacy preferences
- Feedback Prompt: daily dialog for network satisfaction rating (1–5)
- Manual Report Screen: form for submitting network problem reports
- Map Screen: network quality coverage map (Could Have)
- Weekly Report Screen: weekly network experience summary (Could Have)

Hardware Interface

The system interacts with:

- Mobile device sensors
- GPS module
- Internet connection

Software Interface

The application interacts with:

- MySQL database
- Android APIs
- REST APIs
- Backend server
- Android Telephony API: signal strength and cell information
- Android Location API: GPS coordinates
- Android Work Manager / Job Scheduler: background task scheduling
- SQLite Database — local data storage
- RESTful Backend API: data synchronization
- Google Maps API: coverage map visualization

7. Database Requirements

The database stores:

- User information
- QoE feedback
- Network metrics
- Device information
- GPS coordinates
- Timestamps

Main Database Entities

- User
- Feedback
- NetworkMetrics

Requirement	Depends On
FR02	FR01 — Statistics display requires background monitoring to be active
FR03	FR01 — Notifications require background monitoring
FR09	FR08 — Backend sync requires local storage first
FR10	FR07 — Map requires GPS-tagged data to render
FR14	FR11 — QoE feedback depends on the user prompt system
NFR04	FR09 — Security applies to all data transmission
NFR06	FR08 — Offline reliability depends on local storage implementation

- DeviceInformation
- LocationData

8. Dependencies

9. Security Requirements

To ensure privacy and security:

- Data transmission must use encryption
- User authentication must be secure
- Access control mechanisms must be implemented
- Sensitive information must remain confidential
- Only authorized users should access reports

10. Future Enhancements

Possible future improvements include:

- AI-based QoE prediction
- Machine learning analytics
- Cross-platform support (iOS)
- Integration with telecom operator systems
- Real-time outage detection
- Cloud-based analytics dashboards

11. Conclusion

This Software Requirements Specification document defines the functionalities, requirements, constraints, and expected behavior of the proposed QoE mobile application. The document provides a clear understanding of the system for both developers and stakeholders. By combining user feedback with automated network performance monitoring, the application aims to improve network optimization, enhance customer satisfaction, and support better decision-making within the telecommunications sector in Cameroon.